

Anticancer effects of sweet potato protein on human colorectal cancer cells

AIM: To investigate the effects of proteins purified from sweet potato storage roots on human colorectal cancer cell lines. METHODS: 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) assay, Hoechst 33258 nuclear staining and Boyden transwell chamber methods were used to determine whether purified sweet potato protein (SPP) from fresh sweet potato roots affected proliferation, migration and invasion, respectively, of human colorectal cancer SW480 cells in vitro. The inhibitory effects of SPP on growth of human colorectal cancer HCT-8 cells intraperitoneally xenografted in nude mice and spontaneous lung metastasis of murine Lewis lung carcinoma 3LL cells subcutaneously transplanted in C57 BL/6 mice were also investigated in vivo. RESULTS: SPP inhibited the proliferation of SW480 cells in a dose-dependent manner, with an IC₅₀ value of 38.732 μ mol/L ($r^2 = 0.980$, $P = 0.003$) in the MTT assay. Hoechst 33258 nuclear staining further revealed inhibition of cell viability and induction of apoptosis by SPP. The transwell assay disclosed significant reduction in migrated cells/field by 8 μ mol/L SPP (8.4 ± 2.6 vs 23.3 ± 5.4 , $P = 0.031$) and invaded cells/field through the ECMatrix by 0.8 μ mol/L SPP, compared with the control (25.2 ± 5.2 vs 34.8 ± 6.1 , $P = 0.038$). Both intraperitoneal (ip) and intragastric (ig) administration of SPP led to significant suppression of growth of intraperitoneally inoculated HCT-8 cells in nude mice to $58.0\% \pm 5.9\%$ ($P = 0.037$) and $43.5\% \pm 7.1\%$ ($P = 0.004$) of the controls, respectively, after 9 d treatment. Bloody ascites additionally disappeared after ip injection of trypsin inhibitor. Notably, ig and ip administration of SPP induced a significant decrease in spontaneous pulmonary metastatic nodule formation in C57 BL/6 mice (21.0 ± 12.3 and 27.3 ± 12.7 nodules/lung vs 42.5 ± 4.5 nodules/lung in controls, respectively, $P < 0.05$) after 25 d treatment. Moreover, the average weight of primary tumor nodules in the hind leg of mice decreased from 8.2 ± 1.3 g/mice in the control to 6.1 ± 1.4 g/mice in the ip group ($P = 0.035$). CONCLUSION: SPP exerts significant antiproliferative and antimetastatic effects on human colorectal cancer cell lines, both in vitro and in vivo.